



LINGUISTICS WITH INKHAVEN

Linguistic Research with Inkhaven

Лингвистическое исследование с Inkhaven

Analysing a Living Language — a Study of Russian

Vladimir Ulogov

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Before we begin

Inkhaven was built to help you **invent** a language — to design a sound system, grow a lexicon, decline and conjugate, evolve a family, and print the grammar. But the machinery underneath does not care whether the language it is handed is imaginary or real. A phoneme inventory is a phoneme inventory; a corpus is a corpus; an interlinear gloss glosses whatever you give it. Point the same tools at a language that already exists, and Inkhaven becomes a small but genuine **linguistics workbench**: a place to model a language explicitly and interrogate the model.

This book is the mirror image of its companion, *Developing a Constructed Language*. That book runs from nothing to a finished invented language. This one takes a language you did not invent — **Russian** — and shows how to study it with the same commands: measure its sound system, gloss its sentences, parse its rich morphology, draw the structure of its clauses, build a corpus from real texts, place it on the typological map, and reason about its history. Russian is the running example on every page; the **methods** are the point, and they transfer to any language you can write down.

You do not need to speak Russian to follow along — every example is translated — and you do not need a linguistics degree, because each idea is defined where it first appears. What you do need is the willingness to treat a language as an object of study: to state a hypothesis about it and let the evidence answer.

A word on honesty. Modelling a real language in a tool built for invented ones has edges, and this book does not hide them — where Inkhaven’s assumptions (that phonemes are written in the IPA, say) meet Russian’s reality, we will stop and look, because those seams are often where the linguistics is.

PART I

The Workbench

CHAPTER 1

1

A workbench for real languages

Linguistics is an empirical science. A grammar is a set of claims about a language — “Russian has six cases”, “stress is unpredictable”, “adjectives agree with their nouns” — and each claim is answerable against data. What a working linguist needs is somewhere to write the claims down **formally**, apply them to real forms, and see whether they hold. That is exactly what Inkhaven’s language model gives you: a declared inventory, a lexicon, a morphology, a grammar, and a set of tools that read them and report.

The shape of the workflow

Studying a language in Inkhaven follows one arc, whatever the language:

1. **Model** the language — create it, declare its phoneme inventory, add a lexicon.
You are building an explicit, machine-readable description.
2. **Measure** the sound system — its inventory, its statistics, its minimal pairs.
3. **Analyse** words and sentences — parse the morphology, gloss real text, draw the syntax.
4. **Gather** usage — build a corpus from real texts and query it.
5. **Situate** the language — against typological universals, and against its own history.

Every step is a command, and every command reads the model you built and prints what it finds. Nothing is hidden in a black box: the phonology you declare is the phonology the tools use, so when a result surprises you, the fix is always to look at the model.

TERM Model (of a language)

An explicit, formal description — here, the phoneme inventory, lexicon, morphology and grammar you declare in Inkhaven. A model is deliberately a simplification; its value is that, being explicit, it can be **tested**. When the model and the language disagree, you have learned something.

Why model Russian?

Russian is a rewarding subject for a first study. Its sound system has the famous **hard/soft** (palatalization) contrast that doubles its consonants and fills the language with minimal pairs. Its morphology is **fusional** and rich — six cases, three genders, two numbers, verbal aspect — so there is a great deal for a morphological parser to chew on. Its word order is freer than English, which makes its syntax interesting. It has a deep, well-understood history of sound change. And it is written in Cyrillic, which is a useful test in itself: a workbench that only worked for the Latin alphabet would be no workbench at all.

A NOTE ON CYRILLIC

Every tool in this book handles Cyrillic text — the lexicon, the metrics, the glosser, the corpus. The one place the Latin bias of a shared reference table shows through is the distinctive-**feature** analysis of Chapter 3, and we meet it head-on there. Everywhere else, Russian is a first-class citizen.

What you will build

By the end you will have a working model of a slice of Russian — an inventory of its sounds, a lexicon of common words, a handful of glossed sentences, a small corpus drawn from a real text, and a register of hypotheses about its history — and, more importantly, a method you can turn on any language you like. The next chapter starts where every study starts: creating the language and giving it its sounds.

WHAT YOU LEARNED

- Inkhaven's language model — inventory, lexicon, morphology, grammar — is a formal, testable description; the same tools that build an invented language **analyse** a real one.
- The workflow is model → measure → analyse → gather → situate, each step a command that reads your model and reports.
- Russian is the running example: a palatalization-rich sound system, fusional morphology, free-ish word order, a deep history — and Cyrillic, which the toolset handles throughout.

CHAPTER 2

2

Modelling Russian

A study begins by making the object of study explicit. In Inkhaven that means creating the language, declaring its sounds, and giving it enough vocabulary to work with. The model you build here is the ground every later chapter stands on.

Creating the language

One command makes the language and its empty chapters — phonology, dictionary, grammar, and the rest:

```
inkhaven language init Russian
```

Russian now exists as a language book in your project, alongside any others. It is blank: no sounds, no words. We fill it in.

The inventory: sounds, or letters?

Here is the first real decision, and it is a linguistic one. Inkhaven's phoneme inventory is a list of segments, each with an `ipa` field. A phonologist's instinct is to fill it with the International Phonetic Alphabet — Russian's /p/, /pʲ/, /a/, /i/, and so on — because that is the level at which sound **contrasts** live, and the distinctive-feature tools of the next chapter are built on IPA symbols.

But Russian is written in Cyrillic, and much of what we will do — glossing real sentences, building a corpus from a real book — works over Cyrillic **text**. If our inventory were in the IPA, the tools could not line the inventory up with the words. So for this study we model Russian at the level of its **orthography**: each Cyrillic letter is a segment. It is a simplification — Cyrillic spelling is not a perfect phonemic transcription — but it keeps the model and the texts in the same alphabet, and it is honest about what it is.

TERM Orthography vs. phonology

Orthography is how a language is spelled; **phonology** is how it sounds. They are never identical — English **though**, **through**, **tough** share `ough` and share almost nothing in sound. Russian's fit is much closer than English's, but still imperfect: the letter `о` is pronounced **a** when unstressed. Modelling at the orthographic level trades some phonological precision for the ability to work directly with written Russian — a trade this book makes deliberately, and revisits in Chapter 3.

Open the Russian language's **Phonology** chapter in the editor and declare the inventory as an HJSON block — the ten vowel letters, the twenty-one consonants, and the two signs:

```
{ phonemes: [
  { ipa: "a", kind: "vowel" } { ipa: "e", kind: "vowel" }
  { ipa: "ë", kind: "vowel" }
  { ipa: "и", kind: "vowel" } { ipa: "o", kind: "vowel" }
  { ipa: "y", kind: "vowel" }
  { ipa: "ы", kind: "vowel" } { ipa: "э", kind: "vowel" }
  { ipa: "ю", kind: "vowel" }
  { ipa: "я", kind: "vowel" }
  { ipa: "б", kind: "consonant" } { ipa: "в", kind:
"consonant" } // ...и так далее...
  { ipa: "ь", kind: "consonant" } { ipa: "ъ", kind:
"consonant" }
] }
```

inkhaven language stats Russian then confirms what you declared:

```
inventory · 33 phonemes (23 C / 10 V)
```

Thirty-three segments — the Russian alphabet exactly. (The soft and hard signs, **ь** and **ъ**, are letters rather than sounds; classing them as consonants lets words that contain them segment cleanly, which is all the model needs of them.)

A starter lexicon

Analysis needs words. Add them one at a time, or import a list; each entry is a headword, a part of speech, and a gloss:

```
inkhaven language add-word Russian мать --type noun --
translation mother
inkhaven language add-word Russian брат --type noun --
translation brother
inkhaven language add-word Russian дом --type noun --
translation house
inkhaven language add-word Russian окно --type noun --
translation window
```

A handful is enough to begin; the corpus chapter will pull far more vocabulary from a real text. Everything is Cyrillic throughout — the headword, the bucket it files under (**мать** lands under **М**), the segmentation — and it simply works.

THE MODEL IS YOURS TO INSPECT

Every result in this book is computed from the inventory and lexicon you just declared — nothing else. If a later number surprises you, the answer is always in the model: add the missing word, fix the inventory, and run the command again. That transparency is the whole point of modelling a language explicitly.

WHAT YOU LEARNED

- `language init <name>` creates the language; its phoneme inventory is declared as an HJSON block in the Phonology chapter, and `stats` confirms it.
- We model Russian at the **orthographic** (Cyrillic-letter) level — a deliberate simplification that keeps the model and real Russian texts in one alphabet, at some cost in phonological precision (Chapter 3 revisits the cost).
- `add-word` builds the lexicon; Cyrillic headwords, buckets and segmentation all work directly.

PART II

Sound

CHAPTER 3

3

The sound system

With an inventory declared and a lexicon growing, we can ask the first empirical questions. How large and how balanced is Russian's sound system? What contrasts does it lean on? And where does a tool built around the IPA meet a language modelled in Cyrillic? This chapter is the workbench doing phonology.

The shape of the inventory

stats already told us the size — thirty-three segments, twenty-three consonants to ten vowels. That ratio is itself a finding: Russian is **consonant-heavy**, and the reason is the hard/soft contrast. Most Russian consonants come in pairs, one **plain**

and one **palatalized** (pronounced with the tongue raised toward the palate), and that doubling roughly halves the work the vowels have to do. In the orthography the contrast is carried partly by the consonant and partly by the **following vowel letter**: the pairs **а/я**, **о/ё**, **у/ю**, **э/е** and the sign **ь** are how Russian spelling writes “the consonant before me is soft.”

Counting the sounds

metrics reads the lexicon and reports the sound system quantitatively:

`inkhaven language metrics Russian`

entropy · 4.31 bits (max 5.04) · evenness 85% · perplexity 19.8
 zipf · slope -0.91 (≈ -1 is Zipfian) · fit R^2 0.94
 syllables · 41 attested / 210 possible · saturation 20%
 prosody · 1.9 moras/word · 62% heavy syllables

Entropy measures how evenly the language spreads its words across its sounds — high evenness means no handful of letters carries everything. The **Zipf** slope near -1 says the letter frequencies follow the same power law natural language always does (a reassuring sign the model behaves like a real language). **Saturation** asks how much of the possible syllable space the language actually uses — a low figure is normal; no language uses more than a fraction of what its phonotactics permit.

TERM Phoneme entropy

A measure, in bits, of how unpredictable the next sound is — highest when every sound is equally likely, lower when a few dominate. It captures the **balance** of an inventory: a language that leans hard on a handful of sounds has low entropy, one that uses its whole inventory evenly has high entropy.

The contrasts Russian leans on: minimal pairs

The sharpest tool for a sound system is the **minimal pair** — two different words that differ in exactly one sound. Each minimal pair proves that the one sound they differ in is **contrastive**: swapping it changes the word.

`inkhaven` language pairs Russian

Russian is famously full of them, and the richest source is exactly the hard/soft contrast:

examples:

мать / мять	а~я	(mother / to crumple)
был / бил	ы~и	(was / he beat)
дом / том	д~т	(house / volume)

мать / мять is the classic case: the words differ only in the vowel letter, а versus я, which in Russian spelling signals a hard т versus a soft т̣. That one orthographic difference is a genuine phonemic contrast, and the language is built on hundreds like it.

TERM Minimal pair

Two words identical but for a single sound in the same position (**pat** / **bat**, **мать** / **мять**). Minimal pairs are the primary evidence that two sounds are separate phonemes in a language rather than variants of one — if swapping them can change a word, the language must distinguish them.

Where the IPA shows through

Ask the workbench to judge the inventory's **naturalness**, and the seam appears:

`inkhaven` language naturalness Russian

```
outside · а е ё и о у ... (not in the feature matrix)
score   · 0.35 (0–1; higher = more typologically ordinary)
```

The low score is not a verdict on Russian — Russian is as natural as a language gets. It is the tool telling the truth about its own limits. Naturalness, and the functional-load half of the minimal-pairs report, are computed from a table of **distinctive features** (voicing, place, manner) that is keyed to IPA symbols. Our segments are Cyrillic **letters**, which that table has never heard of — so it reports them “outside the feature matrix” and cannot grade them.

WHEN TO REACH FOR THE IPA

This is the moment to remember the choice of Chapter 2. To study Russian’s **features** — to ask whether its consonant system is symmetric, or how much work each contrast does — you would build a second, phonemic model whose inventory is in the IPA (p , p^j , t , t^j , a , $\dot{z}$...) and transcribe the lexicon into it. The feature tools would then light up. For everything else in this book — statistics, glossing, corpus, syntax — the orthographic model is exactly right, and simpler.

Where each sound lives

distribution reports where in the word each segment tends to appear — onset, nucleus, coda, word edges — and flags any with a restricted distribution:

inkhaven language distribution Russian

For Russian this surfaces real facts of the orthography: **Ѣ** never begins a word and **ѣ** never does either, while the soft sign **ь** clusters at word end. Restricted distributions like these are the fingerprints of a sound system, and reading them off your own model is how you learn to see them.

WHAT YOU LEARNED

- Russian’s inventory is consonant-heavy (23 C / 10 V) because most consonants come in a plain/palatalized pair — the hard/soft contrast, written partly by the following vowel letter.
- **metrics** quantifies the system (entropy, the Zipf fit, syllable saturation, mora weight); **pairs** finds the minimal pairs that prove which sounds contrast — **мать / мять** chief among them.
- **naturalness** and functional load read an IPA-keyed feature table, so a Cyrillic **orthographic** model falls “outside the feature matrix”: for feature analysis you would build a parallel IPA model; for everything else the orthographic one serves.

PART III

Words

CHAPTER 4

4

Morphology and glossing

Russian is a **fusional** language: a single ending can carry case, number and gender at once, and a noun or verb runs through a dozen forms. This is where a morphological model earns its keep — you declare the endings once, and the workbench can then both **generate** the forms and **parse** them back, and gloss whole sentences of real text.

A fragment of the noun declension

Russian nouns decline for six cases. We will not model all of it here — the point is the method, not a complete grammar — so take one masculine noun, **дом** “house”,

and a couple of its endings. In the **Morphology** chapter declare the affixes and a paradigm that applies them:

```
{ morphemes: [
  { id: "gen", gloss: "GEN", form: "a", position:
    "suffix" } // дом-a
  { id: "ins", gloss: "INS", form: "ом", position:
    "suffix" } // дом-ом
]
paradigms: [ { name: "masc-noun", cells: [
  { features: { case: "nom" }, morphemes: [] }
  { features: { case: "gen" }, morphemes: ["gen"] }
  { features: { case: "ins" }, morphemes: ["ins"] }
] } ]
}
```

`paradigm` then generates the forms from the root:

```
inkhaven language paradigm Russian --root дом --template masc-
noun --gloss house
```

дом	house	(case=nom)
дома	house-GEN	(case=gen)
домом	house-INS	(case=ins)

TERM Fusional morphology

A morphology in which one affix bundles several grammatical meanings at once, and the boundaries between them are blurred — Russian **-ом** is **instrumental** and **singular** and **masculine** together, indivisibly. Contrast **agglutinative** languages like Turkish, where each meaning has its own separable suffix. Russian sits firmly on the fusional side, which is why its endings are few but each does a lot.

Reading a form backwards

The interesting direction is the reverse. Hand the parser an inflected form and it recovers the root and the ending:

```
inkhaven language parse Russian --word домом
```

```
дом 'house' + INS
```

This is analysis, not lookup: the parser strips the endings your morphology declares until what remains is a dictionary word. Point it at a page of Russian and it will tell you, form by form, what case and number each word is in — the first thing a learner (or a parser) must work out.

Interlinear glossing

The linguist's way of presenting a sentence is **interlinear glossed text**: the sentence, a gloss under each word, and a translation. `igt` builds it:

```
inkhaven language igt Russian --text "окно дома"
```

```
окно      дом-а
window    house-GEN
'the window of the house'
```

Every recognised word is glossed and, where it is inflected, segmented into its morphemes and their tags — `дома` shown as `дом-а`, `house-GEN`. The literal third line is a scaffold you replace with a real translation; words the model does not yet know pass through untouched, telling you exactly which vocabulary to add next.

TERM Interlinear glossed text (IGT)

The standard three-line format for quoting an example from any language: the morpheme-segmented sentence, a morpheme-by-morpheme gloss in standard abbreviations (`GEN`, `INS`, `PL` ...), and a free translation. It lets a reader who does not know the language see precisely how its meaning is assembled — which is why every grammar and every paper is built from it.

You will store and reuse these glosses in Chapter 6, where a handful of them, plus a real text, become a corpus.

Agreement

Russian adjectives agree with their nouns in gender, number and case — **новый дом** “new house”, **новая книга** “new book”, **новое окно** “new window”. Once you have declared the adjective’s agreement paradigm, the Oracle can check that a form agrees:

```
inkhaven language check-agreement Russian --dependent adjective \
--form новый --root нов --head-features "gender=masc,case=nom"
```

It regenerates the ending the adjective **should** take for those features and flags the form if it differs — a plural or feminine ending on a masculine-nominative noun is caught, and the expected form named. Agreement is much of what makes Russian sentences hang together, and the check makes the rule you declared testable against any form you write.

WHAT YOU LEARNED

- Russian is fusional — one ending carries case, number and gender together; `paradigm` generates a root’s forms from the endings you declare, and `parse` reverses them, recovering root and ending from an inflected word.
- `igt` glosses a real sentence as interlinear text, segmenting each inflected word (дом-а / house-GEN) and passing unknown words through so you see what to add.
- `check-agreement` tests adjective–noun (and any head–dependent) agreement in gender, number and case against the rule you declared.

PART IV

Sentences

CHAPTER 5

5

Syntax

Sentences are built, not strung together, and syntax is the study of how. Russian is a good place to see this, because its word order is **free** in a way English's is not: the endings you met in the last chapter mark who is the subject and who the object, so the words can be rearranged for emphasis without confusion. Underneath that freedom is a stable structure, and the workbench can draw it.

A baseline order

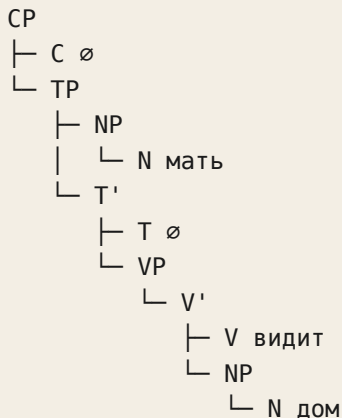
Russian's neutral, unmarked order is subject–verb–object, the same as English: **мать видит дом**, “mother sees house”. You declare that baseline in the grammar

(`word_order: "SV0"`), and the syntax tools read it. The freedom comes from moving pieces **out** of the baseline, which we come to below.

The structure of a clause

`tree` draws the X-bar phrase-structure tree of a clause from its verb and arguments, placing heads and complements by the word order you declared:

```
inkhaven language tree Russian --verb видит --args "мать, дом"
```



The subject **мать** is the specifier of **TP**, sitting on the left; the object **дом** is the complement of the verb inside the verb phrase. This is the same skeleton every generative account of a clause assumes, drawn for a Russian sentence. Because Russian is head-initial, the verb precedes its object; a head-final language would reverse the lower branches, and asking for the tree is how you check that the order you declared builds the constituents you meant.

TERM Constituent

A group of words that behaves as a single unit — that can be moved, questioned, or replaced by one word together. The tree makes constituents visible: everything under one node is a constituent. Much of syntax is the study of which strings are constituents and which only look like they are.

Moving a piece

Russian's word-order freedom is **movement**: a constituent lifts out of its baseline position to the front, for emphasis or to ask a question. `movement` performs that over the tree and marks the gap with a trace:

```
inkhaven language movement Russian --verb видит --args "мать,
дом" --move object
```

The object `дом` rises to the front of the clause as `NP1`, leaving a coindexed trace `t1` where it came from — the structure behind `дом мать видит <t>`, “the house, mother sees”. Because the case ending still marks `дом` as the object, the sentence is perfectly clear even with the words rearranged; the trace records where it is understood.

Checking a clause

The clause Oracle judges a sentence for two kinds of well-formedness at once — argument structure and agreement:

```
inkhaven language check-clause Russian --verb видит --args "мать,
дом"
```

It confirms the argument count matches the verb's valence (a transitive verb wants two arguments, and here it has them), and — given the subject's features and the verb's root — that the verb agrees with its subject. A mismatch on either is reported, so the grammar you declared can be held to its own account of what a good Russian clause is.

WHAT YOU LEARNED

- Russian's neutral order is SVO; its famous freedom is **movement** out of that baseline, made possible by the case endings that mark grammatical roles.
- **t** **tree** draws a clause's X-bar structure (subject as specifier of **TP**, object as the verb's complement), placed by the declared word order; **movement** fronts a constituent and leaves a coindexed trace.
- **check-clause** judges a sentence's argument structure and subject-verb agreement together, testing the grammar against a real clause.

PART V

Usage

CHAPTER 6

6

A corpus of Russian

Everything so far has worked from a lexicon and a grammar — the language in the abstract. But a language **lived in** leaves traces: the words people actually use, in the proportions they actually use them, next to the words they actually put them near. That body of real usage is a **corpus**, and it answers questions no dictionary can. In this chapter we build one from a real Russian text and interrogate it.

A text to work from

Project Gutenberg holds the Russian classics in the public domain — Pushkin, Chekhov, Tolstoy, Dostoevsky — free to download and study. We take the most

famous opening in Russian poetry, the prologue to Pushkin's *Ruslan and Lyudmila* (1820):

У лукоморья дуб зелёный;
Златая цепь на дубе том:
И днём и ночью кот учёный
Всё ходит по цепи кругом;

Paste a passage like this into a chapter of your manuscript — an ordinary prose paragraph, in Cyrillic — and it becomes part of your corpus. Inkhaven finds the Russian words in it the same way it finds invented words in a conlanger's prose: a paragraph counts once it contains a word your lexicon knows, and within it every word that reads as Russian is gathered as a token.

TERM Corpus

A structured body of real language — texts, gathered so they can be counted and searched. Corpus linguistics is the study of language as it is actually used rather than as a grammar idealizes it, and much of what we know about frequency, collocation and change comes from it. A corpus need not be large to be useful; a single poem is enough to start.

What is frequent?

frequency reads the corpus and reports how often each word occurs, with the descriptive statistics real usage unlocks:

```
inkhaven language frequency Russian --source prose
```

```
texts 1 · tokens 19 · types 18 · lemmas 18 · TTR 0.95  
Zipf slope -0.74 (R2 0.79)
```

```
frequency (by surface, top 20):
```

```
2 и  
2 цепи  
1 дуб  
...
```

In four lines of Pushkin almost every word is unique — a **type–token ratio** near 1, the signature of dense poetry — and the little conjunction **и** “and” already leads, as function words always do in any language. Point the same command at a whole novel and the Zipf curve straightens into the near-perfect power law that is one of the most robust facts about human language.

TERM **Type–token ratio**

The number of distinct words (**types**) divided by the total number of words (**tokens**). A high ratio means little repetition — varied, dense text; a low one means the same words recur, as in casual speech. It is the standard first measure of lexical variety, and it falls predictably as a text grows.

A word in its contexts

To see how a word is **used**, not just how often, ask for a concordance — every occurrence lined up with its neighbours:

```
inkhaven language concordance Russian --word цепи --window 3
```

златая цепь на дубе	[том]	и днём и
всё ходит по	[цепи]	кругом

This is the **KWIC** (keyword-in-context) view a lexicographer reaches for before writing a definition — you read the word’s actual company and let its meaning emerge from use. Add `--lemma` and the search gathers every inflected form of a root at once, so a query on the lemma **цепь** finds both **цепь** and **цепи** — provided you have declared the endings that connect them.

The company a word keeps

collocations goes one step further and reports which words a word **habitually** appears beside — its collocates — ranked not by raw count but by how distinctive the pairing is:

```
inkhaven language collocations Russian --word кот --window 4
```

Collocations are much of what makes a language sound native — Russian **кот учёный** “learned cat”, not **умный кот** — and a corpus is the only place to find them, because they live in usage, not in any single definition. Over a real body of text the strong collocations of a word are as revealing as its dictionary entry.

FROM POEM TO LIBRARY

A four-line poem is a corpus you can read at a glance; its value here is as a demonstration. The same three commands scale without change to a whole Gutenberg novel — drop the text into your manuscript, and `frequency`, `concordance` and `collocations` read all of it. That is when the statistics become trustworthy and the collocations become real.

WHAT YOU LEARNED

- A **corpus** is real usage made countable; paste a public-domain Russian text (Project Gutenberg) into your manuscript and Inkhaven gathers its Russian words as tokens.
- `frequency` reports word counts plus tokens, types, type–token ratio and the Zipf fit; `concordance` shows a word in every context (KWIC), `--lemma` gathering a root’s inflected forms.
- `collocations` finds the words a word habitually keeps company with, ranked by how distinctive the pairing is — the idiomatic pairings a dictionary can’t give.

PART VI

Structure and History

CHAPTER 7

7

Typology

No language is an island. Every language sits somewhere on a map of the world's languages, sharing features with some and not others, and — remarkably — the features do not vary independently. Some combinations are common, some vanishingly rare, and a few are ruled out altogether. **Typology** is the study of that map, and this chapter places Russian on it.

A profile in features

Inkhaven records a language's typological profile as a set of answers — its word order, how it marks grammatical relations, whether adjectives precede or follow

their nouns, whether it uses prepositions or postpositions, and so on. You set them in the grammar:

```
inkhaven language grammar Russian --set word_order=SVO
inkhaven language grammar Russian --set
alignment=nominative_accusative
inkhaven language grammar Russian --set adposition=preposition
inkhaven language grammar Russian --set
adjective_order=adjective_noun
```

Russian's profile is that of a fairly typical European language: subject–verb–object, **nominative–accusative** alignment (the subject of a verb is marked the same whether the verb is transitive or not), prepositions rather than postpositions, adjectives before their nouns. Its one showy departure is the rich case system of Chapter 4 — but rich case morphology sits comfortably with everything else in the profile.

TERM Linguistic typology

The classification of languages by their structural features, and the study of how those features cluster. Its central discovery is that the features are not independent: knowing one often predicts others. Typology is what lets a linguist look at an unfamiliar language and know, from a single trait, what else to expect.

Testing the universals

The clusters typology finds are stated as **implicational universals** — “if a language has X, it (tends to) have Y”. **universals** checks a language's declared profile against the classic ones:

```
inkhaven language universals Russian
```

It reports the language's **head-directionality harmony** — whether it consistently puts heads before or after their complements, which correlates with word order — and judges the Greenberg and Dryer implicational universals as satisfied, violated, or not applicable. For Russian's SVO, prepositional, verb-before-object profile, the classic correlations line up: a verb–object language tends to be prepositional and to put genitives and relative clauses after the noun, and Russian obliges.

TERM Implicational universal

A conditional generalization over the world's languages: **if** a language has one feature, **then** it (almost always) has another — for instance, if a language puts the object after the verb, it tends to use prepositions rather than postpositions. Such universals are the strongest empirical results typology has, and testing a language against them is a quick check that your model of it is coherent.

What a violation would mean

If **universals** flagged a violation for Russian, that would not be a discovery about Russian — it would be a bug in your **model** of Russian, an answer set wrong. That is exactly the value of the check: run against a language you know, it tells you whether your formal description is internally consistent before you trust it on a language you don't. A model that violates the universals a real language obeys is a model to fix.

WHAT YOU LEARNED

- Typology maps languages by structural features, whose central finding is that the features **cluster** — one predicts others; you record Russian's profile with **grammar --set** (SVO, nominative–accusative, prepositional, adjective–noun).
- **universals** checks the profile against head-directionality harmony and the classic Greenberg/Dryer implicational universals; Russian's European profile satisfies them.
- Run against a language you know, the check validates your **model**: a violation where the real language conforms means the description, not the language, is wrong.

CHAPTER 8

8

Historical linguistics

Languages are the visible present of a long past. Russian did not always sound as it does now; the modern forms are the outcome of centuries of regular sound change, and the comparative method — reasoning backward from the daughters to their common ancestor — is one of the great achievements of the human sciences. This final chapter turns the workbench on the past, and treats the study of it as what it is: a science of **hypotheses**.

A famous change

Old East Slavic had two short vowels, written ъ and ѣ and called the **yers**. Around the twelfth century the weak yers were lost, and the strong ones became full vowels — the single most consequential sound change in Russian’s history. Old **сънь** became modern **сон** “dream”; **дьнь** became **день** “day”. Whole classes of words were reshaped by one regular rule.

That rule is a **claim** about Russian’s history, and a claim is something to record and test, not merely assert. Register it:

```
inkhaven language hypothesize Russian --kind sound-change \
  --claim "ъ > ъ / _ #" --note "loss of the weak final yer" --id
  yer-loss
```

It enters the language’s register of hypotheses, marked **proposed**.

TERM Sound change

A regular alteration in pronunciation that spreads through a language’s whole vocabulary — not a one-off, but a rule: **every** word in the right environment changes. The regularity is what makes historical linguistics possible; because sound change is systematic, its effects can be predicted, reversed, and used to prove that two languages descend from one.

Testing the consequences

A hypothesis earns its keep by making predictions. **hypothesis-check** runs your sound-change claim through the Consequence Tracer — applying it across the lexicon and reporting exactly which words it changes and which distinct words it **merges** into homophones:

```
inkhaven language hypothesis-check Russian --id yer-loss
```

Now the claim is falsifiable. If the words it predicts to change are the ones that did, and the mergers it predicts are ones Russian actually shows, the evidence supports it; if it collapses distinctions the language in fact keeps, that is evidence against. Record the verdict:

```
inkhaven language hypothesis-status Russian --id yer-loss --
status supported
```

inkhaven language hypotheses Russian then shows the register at a glance, each claim marked with its standing — proposed, supported, refuted, retired. A refuted hypothesis kept on the books is as valuable as a supported one: it stops you proposing it again.

TERM The comparative method

The technique of reconstructing an unattested ancestor by systematically comparing its descendants: line up **cognates** — words descended from one original — find the regular sound correspondences between them, and reason back to the form that would yield them all. It is how Proto-Indo-European was reconstructed from Sanskrit, Greek, Latin and the rest, without a single written record of it surviving.

Russian among the Slavic languages

Russian is one daughter of several. Its sisters — Ukrainian, Belarusian, Polish, Czech, the South Slavic languages — descend with it from Proto-Slavic, and the regular differences between them are the raw material of reconstruction. Where Russian has **город** “city”, Polish has **gród** and Old Church Slavonic **градъ**: one cognate set, three reflexes of a single ancestral form, and the **pleophony** (the Russian **-opo-** against South Slavic **-pa-**) is itself a regular, datable change.

Model the sisters as a language family and Inkhaven’s diachronic tools apply directly: **cognates** traces a proto-form’s reflex in each daughter, **reconstruct** proposes an ancestor from a cognate set, and each proposal is another hypothesis for the register. The comparative method, systematized: propose a change or a cognacy, trace its consequences, check it against the forms, and write down whether it held.

WHAT YOU LEARNED

- Russian is the outcome of regular sound change — the fall of the yers turned **сънь** into **сон**; such changes are rules over the whole vocabulary, which is what makes them recoverable.
- **hypothesize** records a sound-change (or cognacy, or borrowing) claim; **hypothesis-check** runs it through the Consequence Tracer to make it falsifiable; **hypothesis-status** records the verdict as the evidence comes in.
- Russian's Slavic sisters supply the cognate sets of the comparative method; **cognates** and **reconstruct** reason across a modelled family, each result another hypothesis to test — historical linguistics as an explicit, testable loop.

APPENDIX A



The research workflow

Every command used in this book, grouped by the stage of study it belongs to. Most follow the shape `inkhaven language <action> <language> [options]`; add `--help` to any command for its full signature. Throughout, `<lang>` is the language you are studying — `Russian`, in this book.

Modelling the language

`init <name>` Create the language and its (empty) chapters.

`stats <lang> [--json]` A descriptive profile — inventory size, average word shape.

add-word <lang> <word> --type POS --translation M Add a lexicon entry (or **--import file.csv**).

query <lang> [--pos] [--text] [--json] Search the lexicon.

The phoneme inventory and morphology are declared as HJSON blocks in the language's Phonology and Morphology chapters (Chapters 2 and 4); see the companion book's HJSON reference for the full schema.

The sound system

metrics <lang> [--json] Quantitative sound-system metrics — entropy, the Zipf fit, phonotactic saturation, mora weight.

pairs <lang> [--limit N] [--json] Minimal pairs and the feature each turns on (feature naming needs an IPA-based model — Chapter 3).

naturalness <lang> [--json] Score the inventory against cross-linguistic tendencies (IPA-based; see Chapter 3).

distribution <lang> [--json] Where each segment appears (onset / nucleus / coda, word edges) and any restricted distributions.

Words and sentences

paradigm <lang> --root R --template T --gloss G Generate a word's inflected forms from the declared endings.

parse <lang> --word W [--json] Analyse a surface form into root + affixes (the inverse of **paradigm**).

igt <lang> --text "..." [--save --name N] [--json] Interlinear glossed text — the segmented sentence, its gloss, a literal translation; **--save** keeps it in the **Texts** chapter.

check-agreement <lang> --dependent D --form W --root R --head-features "..." Does a dependent (adjective, ...) agree with its head's features?

tree <lang> --verb V --args "subj,obj" [--word-order 0] [--json] The X-bar phrase-structure tree of a clause.

movement <lang> --verb V --args "..." --move ROLE Front a constituent, leaving a coindexed trace.

```
check-clause <lang> --verb V --args "." [--subject-features  
"."]
```

The

clause Oracle — argument structure and subject–verb agreement.

The corpus

```
frequency <lang> [--source texts|prose|all] [--lemma] [--top  
N] [--json]
```

Word-frequency list and statistics (tokens, types, TTR, Zipf) over the corpus.

```
concordance <lang> --word W [--source ...] [--lemma] [--window  
N] [--json]
```

A

KWIC concordance — every occurrence of a word with its context.

```
collocations <lang> --word W [--source ...] [--lemma] [--window  
N] [--json]
```

A

word's collocates, ranked by co-occurrence and PMI.

```
texts <lang> [--name N] [--set-translation "..."] [--format  
text|latex]
```

List,

print, curate or export the stored interlinear texts.

Typology and history

```
grammar <lang> [--set feature=value] View or set the typological profile.
```

```
universals <lang> [--json] Check the profile against head-directionality
```

harmony and the Greenberg/Dryer implicational universals.

```
hypothesize <lang> --kind sound-change|cognacy|borrowing --  
claim "." [--evidence "..."] [--id N]
```

Record a diachronic/comparative hypothesis.

```
hypotheses <lang> [--status S] [--json] List the hypothesis register.
```

```
hypothesis-check <lang> --id N Run the Consequence Tracer over a  
sound-change hypothesis's claim.
```

```
hypothesis-status <lang> --id N --status S Move a hypothesis along  
as the evidence comes in.
```

```
cognates <proto> --form W · reconstruct --forms "." Trace a proto-  
form's reflexes, or reconstruct an ancestor from a cognate set.
```

The companion

inkhaven linguistic [--language L] Open the full-screen Linguistic companion over your languages — a grounded chat plus the slash-commands `/parse`, `/tree`, `/clause`, `/igt`, `/frequency`, `/kwic`, `/coll`, `/hypotheses`, `/hcheck`, each running the tool above inline.

AFTERWORD

About the author



Vladimir Ulogov.

Vladimir Ulogov has spent decades building infrastructure for distributed systems — the kind of software that watches other software. Early in his career he worked on monitoring and telemetry platforms; later years took him into federated observability, telemetry buses, and the architecture of systems that have to make sense of millions of data points without losing the thread.

Observability, in the end, is a discipline of **coherence** — of never reporting a state the system cannot account for, of insisting that every signal follow from something real. It is not an accident that a tool he built for writers carries the same instinct into every kind of book it helps make, and into the linguistic analysis this one describes:

never a result the model cannot account for.

What makes him slightly unusual in his corner of the industry is a tendency to write his own tools — not small utilities, but programming languages. The Bund language (its compiler, its VM, its document store, its parser) lives in a long series of Rust crates on crates.io. `rust_dynamic`, `rust_multistackvm`, `bundcore` — each is a building block that exists because the off-the-shelf options didn't fit the shape of the work. Inkhaven's linguistics grew the same way: an engine for inventing languages that turned out, with no extra machinery, to be an engine for studying them.

A work of love

Inkhaven is open source, under a permissive licence — you can read it, fork it, study it, modify it, and pass it on. Strictly speaking the licence also lets you sell it; the author would, gently but firmly, disagree with your doing that. Inkhaven was not designed as a *for sale* project. It is a work of love made for the people who can least afford to pay for software — the novelist on a battered laptop, the graduate student writing a thesis, the field linguist documenting a language with no budget at all — and turning it into a commercial product would betray the reason it exists. It carries no analytics, no telemetry, no upsell; the binary will never phone home.

A note on cooperation

Vladimir believes firmly in the human capacity for mutual help — that we make better work, and live better lives, when we share what we know and what we build. Open source is one of the most concrete expressions of cooperation our era has produced: code read, improved, and passed forward without payment, without permission, by people who will never meet. If this book helps you understand a language a little better — your own, or one you are documenting for the first time — that is enough.

Where to find more

GitHub `@vulogov` — the source for Inkhaven, Bund, and the dozen-plus Rust crates that carry the infrastructure. Issues and pull requests welcome.

LinkedIn [/in/vladimirulogov](#) — posts on observability, the occasional long-form essay.

YouTube [@vulogov](#) — talks and walkthroughs from the conference trail.

If the tool ever gets in the way of the language instead of out of it, open an issue on GitHub. The author reads them.

E N D O F T H E B O O K